

SILVIO CHRISTIAN JOE

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PROFESSIONAL SUMMARY

Informatics Engineering undergraduate at Universitas Dian Nuswantoro (UDINUS), specializing in Natural Language Processing, Tabular Data Analysis, and Generative AI engineering. Experienced in building end-to-end solutions — from training and fine-tuning machine learning models to deploying production-grade APIs and orchestrating autonomous LLM agent pipelines. Certified by Oracle and IBM in AI and Machine Learning. Actively exploring LangGraph, DSPy, and LiteLLM to stay at the frontier of AI engineering.

EDUCATION

Bachelor of Informatics Engineering — Universitas Dian Nuswantoro (UDINUS), Semarang *Aug 2023 – Present*

- Specializing in Data Science: NLP, Tabular Data, and Machine Learning.
- Independently built 30+ AI/ML projects spanning GenAI agents, MLOps, automation pipelines, and predictive modeling.

High School Diploma — Science Track — SMA Kristen YSKI, Semarang

Jul 2020 – May 2023

TECHNICAL SKILLS

Core Focus: NLP, Tabular Data Analysis, Text Mining, Sentiment Analysis

Generative AI: LangChain, LangGraph, CrewAI, RAG, Vector Search, Prompt Engineering

Programming: Python, SQL, Bash

MLOps & DevOps: Docker, Prefect, Git / GitHub, Oracle Cloud (OCI)

Data & Visualization: Pandas, NumPy, Scikit-Learn, TensorFlow, Matplotlib, Seaborn, Tableau, Power BI, Looker Studio

Exploring Next: LangGraph (advanced), DSPy, LiteLLM, Langfuse, Phoenix Arize, FastMCP

Machine Learning: Classification, Regression, Model Fine-tuning, Class Imbalance Handling (SMOTE / SMOTENC)

LLM Providers: Google Gemini, Groq, Hugging Face, OpenAI-compatible APIs

APIs & Frameworks: FastAPI, Flask, Streamlit, python-telegram-bot (PTB)

Vector Stores: FAISS, Pinecone, Oracle AI Vector Search

Databases: MySQL, Google Sheets API

Soft Skills: Analytical Thinking, Problem Solving, Technical Documentation, Self-directed Learning

PROJECTS

Project	Tech Stack	Description
Generative AI & Agentic Applications		
InsightSQL — LangGraph	LangGraph, Gemini 2.5 Flash	Text-to-SQL agent built on cyclic graph reasoning. Queries, validates, and self-corrects SQL errors in real-time with full step-by-step transparency.
InsightData — AI Analyst	Pandas Agent, Gemini 2.5 Flash	ReAct agent that analyzes CSV, Excel, and Google Sheets: auto-generates Python code for data cleaning, statistical analysis, and visualization.
Auto-README Generator	CrewAI, Gemini 2.5 Flash, Streamlit	Multi-agent system that reads Python and Jupyter codebases and generates complete, structured Markdown documentation.
DocuTalk-AI	LangChain, FAISS, Gemini 2.5 Flash	Agentic RAG assistant that routes queries between local document retrieval (PDF, CSV, TXT) and live web search, with incremental indexing.

Portfolio RAG Assistant	<i>LangChain, Gemini 2.5, FAISS</i>	Personalized RAG chatbot that answers recruiter questions about the portfolio; features strict anti-hallucination and bilingual (EN/ID) support.
Backend, APIs & MLOps		
Stuntify API	<i>FastAPI, Scikit-Learn, Pydantic</i>	Production inference API for child stunting risk prediction with strict type coercion, multi-artifact orchestration, and auto-generated Swagger docs.
Insightify Sentiment API	<i>FastAPI, RoBERTa, Transformers</i>	Dual-language (EN & ID) NLP microservice for sentiment analysis. Supports batch processing via Excel/CSV upload and N-gram keyword extraction.
Resume Scanner API	<i>FastAPI, SBERT, TF-IDF</i>	Dual-engine ATS tool: keyword-based strict matching (TF-IDF) and semantic similarity scoring (SBERT) to evaluate resume–job description fit.
Automation & Orchestration		
NovaCal AI Suite	<i>LangChain, Google Calendar API, Gemini Flash</i>	Three-variant Telegram bot suite (Stateful / Ephemeral / Stateless) for full Google Calendar CRUD via natural language conversation.
NovaMail AI	<i>Gmail API, Gemini Flash, Telegram</i>	Autonomous Gmail bot that reads incoming emails, generates context-aware replies, threads them, and sends a summary to Telegram.
Daily AI News Digest	<i>Prefect, DuckDuckGo, Gemini Flash</i>	Fully automated pipeline that scrapes real-time news, summarizes with Gemini, and delivers a daily briefing to Telegram on a schedule.
Machine Learning & Data Science		
Diabetes Prediction	<i>Scikit-Learn, SMOTE, Hyperparameter Tuning</i>	Classification model built with zero data leakage: scaling and SMOTE applied strictly after train/test split, with extensive hyperparameter search.
Depression Predictor	<i>Scikit-Learn, ImbPipeline, Streamlit</i>	End-to-end ML web app demonstrating class-imbalance handling with SMOTE inside a leakage-free ImbPipeline, deployed with a Streamlit UI.
SpendSense	<i>Streamlit, LangChain, Gemini Vision</i>	Personal finance app combining Pandas Agent for natural language queries, Gemini Vision for receipt OCR, and automated Google Sheets ETL.
Data Analytics & Visualization		
Supermarket Sales Dashboard	<i>Tableau Public</i>	Operations dashboard tracking total sales, COGS, and customer ratings across cities, broken down by payment method and product line.
Bike Sales Dashboard	<i>Tableau Public</i>	Executive-level revenue tracker analyzing performance by US state and brand, with a top-10 high-value customer view.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate	— Oracle	2025
Oracle AI Vector Search Certified Professional	— Oracle	2025
Supervised Machine Learning: Classification	— Coursera (IBM)	2025
Supervised Machine Learning: Regression	— Coursera (IBM)	2025
Artificial Intelligence Essentials V2	— IBM	2025

LANGUAGES

Indonesian	Native speaker
English	Professional working proficiency — technical communication and documentation